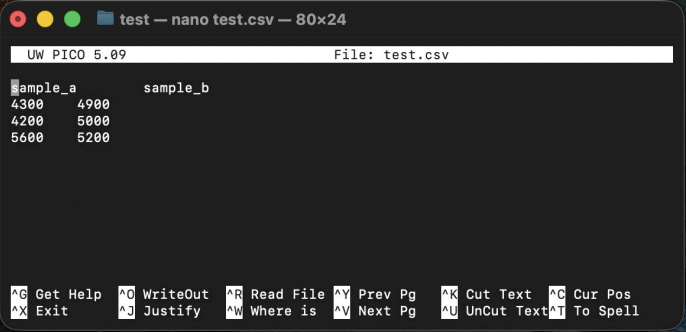


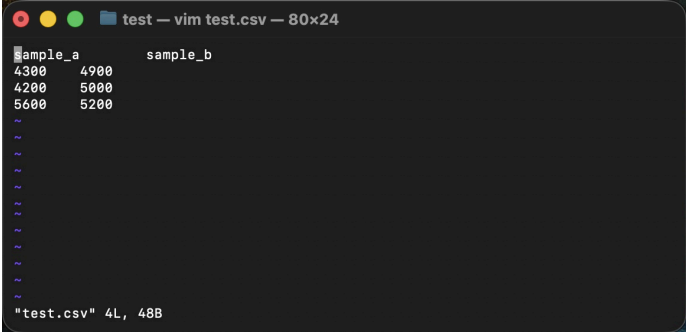
Nano and Vim for Bioinformatics

A practical 3-page guide for editing scripts, configs and cluster jobs safely.

Why this matters: on an HPC cluster you often cannot open VS Code or Excel. You need a terminal editor to fix scripts, sample sheets and pipeline config files directly where the data lives.



Nano: direct editing with shortcut hints at the bottom.



Vim: modal editing with line numbers and command power.

Which editor should I use?

Use nano when...	Use Vim when...
New or under pressure	Need fast search
One small fix	Edit scripts daily
Want visible hints	Need replace/diff
Teaching first survival	Building senior fluency

Mental model

nano
No modes: type, save, exit.

vim
Modes: i to type, Esc to command.

raw data
Do not manually edit FASTQ, BAM or VCF.

Nano survival workflow

Task	Command
Open a file	nano trinity.sh
Save	Ctrl + O, then Enter
Exit	Ctrl + X
Search	Ctrl + W
Cut / paste line	Ctrl + K / Ctrl + U
Line numbers	nano -l samplesheet.csv

Teaching tip: explain that the ^ symbol in nano means Ctrl. So ^O means Ctrl + O, not Shift + 6 then O.

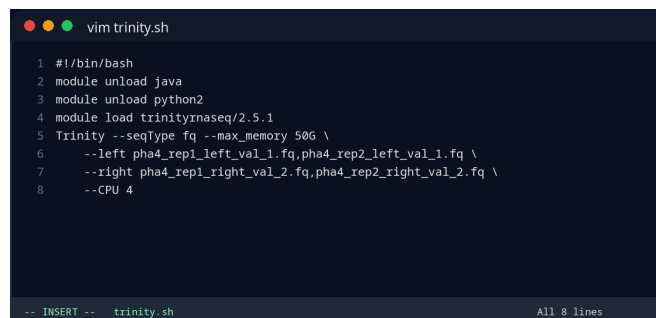
Vim: from panic to practical power

The minimum command set for editing bioinformatics scripts on a server.

Vim has modes

Mode	Use it for	Key
Normal	Move, delete, search, command	Esc
Insert	Actually type text	i
Visual	Select text	v or V
Command	Save, quit, replace	:

If trapped in Vim: press Esc, then type :q! and press Enter. This quits without saving.



```
vim trinity.sh
1 #!/bin/bash
2 module unload java
3 module unload python2
4 module load trinityrnaseq/2.5.1
5 Trinity --seqType fq --max_memory 50G \
6   --left pha4_rep1_left_val_1.fq,pha4_rep2_left_val_1.fq \
7   --right pha4_rep1_right_val_2.fq,pha4_rep2_right_val_2.fq \
8   --CPU 4
-- INSERT -- trinity.sh All 8 lines
```

Typical use: create a shell script directly on the cluster.

The Vim workflow you should memorise

```
vim trinity.sh
i
# edit the file
Esc
:wq
```

Movement and search

Need	Vim command
Top / bottom	gg / G
Go to line 17	17G
Search forward	/text
Next / previous	n / N
Line numbers	:set number

Editing commands

Need	Vim command
Insert text	i
Delete line	dd
Copy / paste line	yy / p
Undo / redo	u / Ctrl+r
Save / quit	:w / :q

Where Vim becomes powerful: replace paths or sample names

```
:%s/raw_data/data/g # replace in whole file
:s/raw_data/data/gc # replace with confirmation
```

In real pipelines this is useful for changing paths inside config.yaml, Snakefile, nextflow.config or run_pipeline.sh.

Senior habit: before risky edits run cp file file.bak or use Git. After editing, run git diff or cat -n file to inspect what changed.

Workshop example: prepare a Trinity assembly job

A realistic use of Vim in RNA-seq assembly teaching.

1. Create and edit the job script

```
vim trinity.sh
```

Press **i** to enter Insert mode, then paste or type the script. Press **Esc**, type **:wq** and press **Enter** to save and quit.

trinity.sh

```
#!/bin/bash
module unload java
module unload python2
module load trinityrnaseq/2.5.1

Trinity --seqType fq --max_memory 50G \
  --left pha4_rep1_left_val_1.fq,pha4_rep2_left_val_1.fq \
  --right pha4_rep1_right_val_2.fq,pha4_rep2_right_val_2.fq \
  --CPU 4
```

Important teaching point: this script tells Trinity it may use 50G, but the sbatch command requests 40G. In production, keep these consistent. Either request at least 50G from SLURM or lower Trinity to match the allocated memory, depending on your cluster rules.

After saving, check the file

```
cat -n trinity.sh
ls -lh pha4_rep*_val_*.fq
bash -n trinity.sh
sbatch -c4 -n1 --mem=40G trinity.sh
queue -u $USER
```

Use **cat -n** to inspect line numbers and check that line continuations end with a backslash where expected.

2. Submit to the cluster

```
sbatch -c4 -n1 --mem=40G trinity.sh
```

This asks SLURM for 4 CPU cores, 1 task and 40 GB memory, then runs the script.

Common mistakes to teach

Mistake	Why it hurts
Forgetting Esc before :wq	You type :wq into the file.
Broken backslash	Shell command splits incorrectly.
Spaces in FASTQ list	Trinity may parse inputs wrongly.
Memory mismatch	Job can be killed by scheduler.
Editing raw FASTQ	Destroys reproducibility.

Senior bioinformatics habits

Habit	Command or behaviour
Make risky edits reversible	cp trinity.sh trinity.sh.bak or git commit first
Check scripts before submitting	bash -n trinity.sh
Record software environment	module list > module_versions.txt
Inspect logs after submission	less slurm-JOBID.out

Bottom line: nano gets beginners unstuck. Vim helps you work quickly, safely and confidently on remote bioinformatics systems.